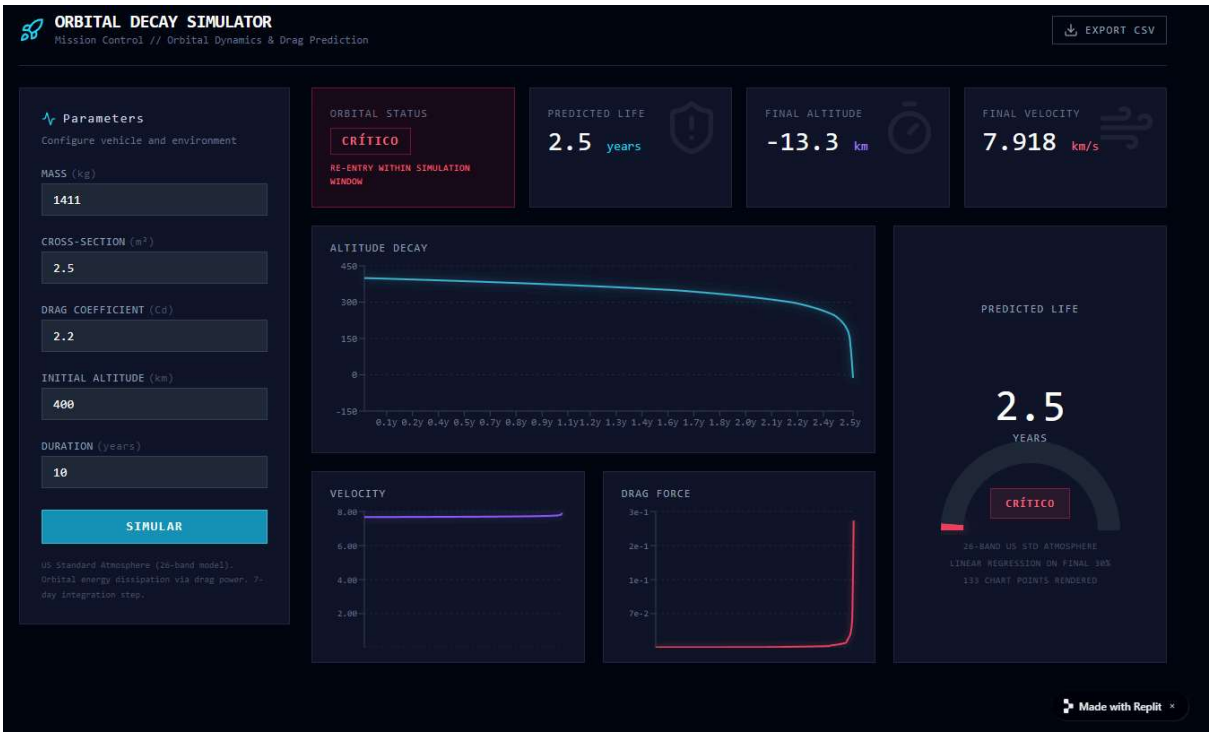
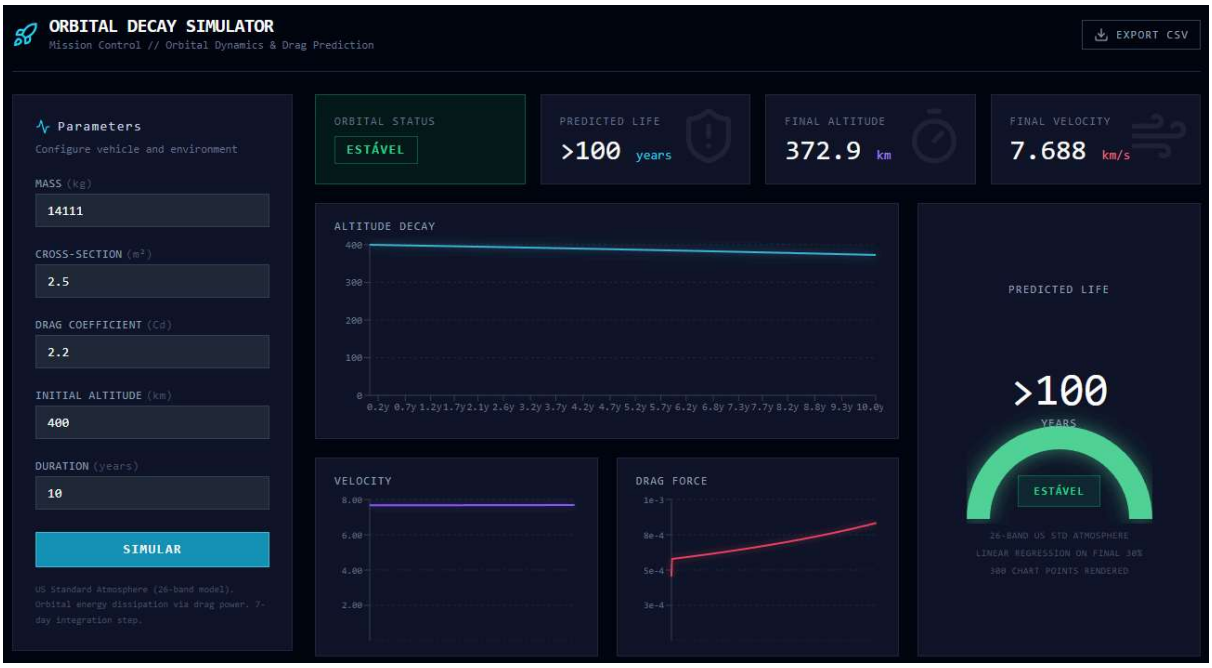
ORBITAL DECAY SIMULATOR
Mission Control // Orbital Dynamics & Drag Prediction
EXPORT CSV
Parameters
Configure vehicle and environment
MASS (kg)
5000
CROSS-SECTION (m²)
2.5
DRAG COEFFICIENT (Cd)
2.2
INITIAL ALTITUDE (km)
400
DURATION (years)
10
SIMULAR
US Standard Atmosphere (26-band model). Orbital energy dissipation via drag power. 7-day integration step.
ORBITAL STATUS
INSTÁVEL
RE-ENTRY WITHIN SIMULATION WINDOW
PREDICTED LIFE
8.8 years
FINAL ALTITUDE
-65.5 km
FINAL VELOCITY
7.951 km/s
ALTITUDE DECAY
VELOCITY
DRAG FORCE
PREDICTED LIFE
8.8
YEARS
INSTÁVEL
26-BAND US STD ATMOSPHERE
LINEAR REGRESSION ON FINAL 30%
300 CHART POINTS RENDERED
Made with Replit


ORBITAL DECAY SIMULATOR
Mission Control // Orbital Dynamics & Drag Prediction
EXPORT CSV
Parameters
Configure vehicle and environment
MASS (kg)
1411
CROSS-SECTION (m²)
2.5
DRAG COEFFICIENT (Cd)
2.2
INITIAL ALTITUDE (km)
400
DURATION (years)
10
SIMULAR
US Standard Atmosphere (26-band model). Orbital energy dissipation via drag power. 7-day integration step.
ORBITAL STATUS
CRÍTICO
RE-ENTRY WITHIN SIMULATION WINDOW
PREDICTED LIFE
2.5 years
FINAL ALTITUDE
-13.3 km
FINAL VELOCITY
7.918 km/s
ALTITUDE DECAY
VELOCITY
DRAG FORCE
PREDICTED LIFE
2.5
YEARS
CRÍTICO
26-BAND US STD ATMOSPHERE
LINEAR REGRESSION ON FINAL 30%
133 CHART POINTS RENDERED
Made with Replit


ORBITAL DECAY SIMULATOR
Mission Control // Orbital Dynamics & Drag Prediction
EXPORT CSV
Parameters
Configure vehicle and environment
MASS (kg)
14111
CROSS-SECTION (m²)
2.5
DRAG COEFFICIENT (Cd)
2.2
INITIAL ALTITUDE (km)
400
DURATION (years)
10
SIMULAR
US Standard Atmosphere (26-band model).
Orbital energy dissipation via drag power. 7-day integration step.
ORBITAL STATUS
ESTÁVEL
PREDICTED LIFE
>100 years
FINAL ALTITUDE
372.9 km
FINAL VELOCITY
7.688 km/s
ALTITUDE DECAY
400
300
200
100
0
0.2y 0.7y 1.2y 1.7y 2.1y 2.6y 3.2y 3.7y 4.2y 4.7y 5.2y 5.7y 6.2y 6.8y 7.3y 7.7y 8.2y 8.8y 9.3y 10.0y
VELOCITY
8.00
6.00
4.00
2.00
DRAG FORCE
1e-3
8e-4
5e-4
3e-4
PREDICTED LIFE
>100
YEARS
ESTÁVEL
26-BAND US STD ATMOSPHERE
LINEAR REGRESSION ON FINAL 30%
300 CHART POINTS RENDERED